Processor



processor.kicad_sch

Power



power.kicad_sch

Interfaces



interfaces.kicad_sch

ASTER2350 | REV A
80 x 65 mm, 4-layer development board
USB-C device + LiPo power path + Wi-Fi and Bluetooth LE
All parts populated; top-side SMT plus factory-soldered THT headers and battery connector.

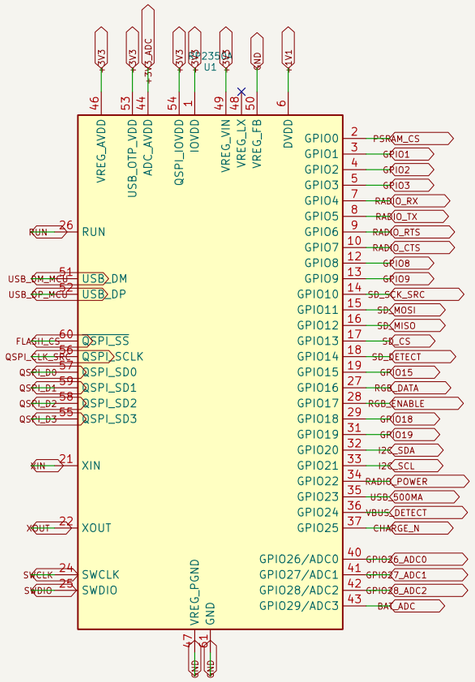
Net labels are global across the three subsystem sheets.
Read README.md and fab/ASSEMBLY.md before ordering.
Prototype: no physical hardware validation or regulatory certification performed.

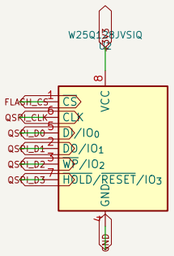
RP2350A / 16 MiB flash / 8 MiB PSRAM

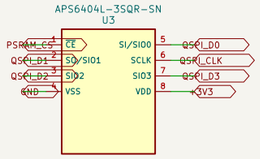
Sheet: /
File: Aster2350.kicad_sch

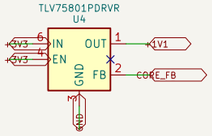
Title: Aster2350 development board

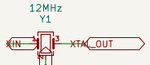
Size: A4	Date: 2026-11-09	**Rev: A**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 1/4

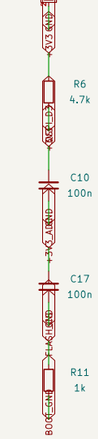


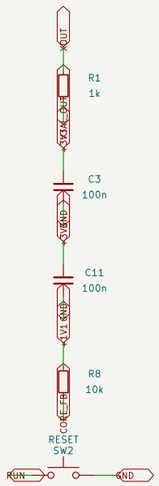




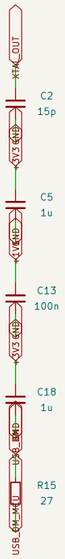








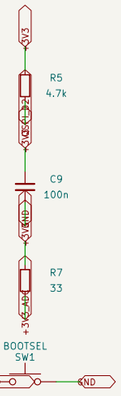












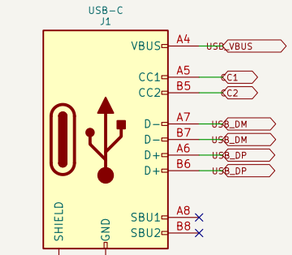
EXTERNAL DVDD: 1.1 V from U4, VREG_VIN and VREG_AVDD remain powered; VREG_FB=GND; LX=NC.
Firmware must set VREG.HIZ. PSRAM uses QMI CS1n on GPIO0. BOOTSEL only shorts flash CS through 1k.

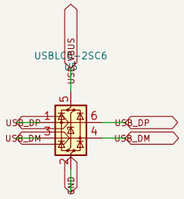
Autonomous engineering prototype

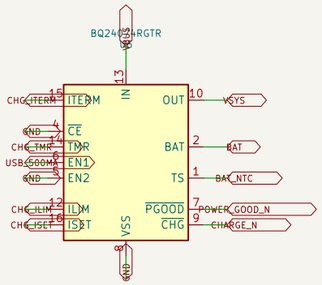
Sheet: /Processor/
File: processor.kicad_sch

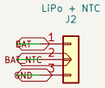
Title: Aster2350 — Processor, QSPI memories and external core supply

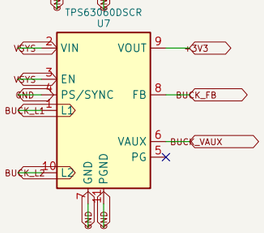
Size: A3	Date: 2026-11-09	**Rev: A**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 2/4

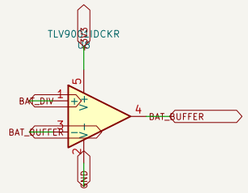


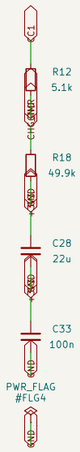


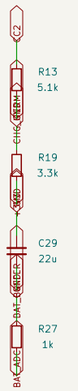




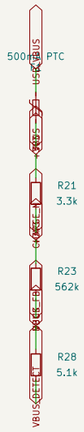


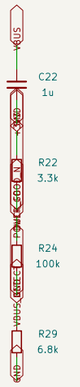




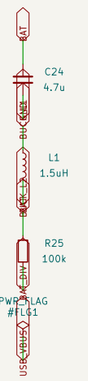




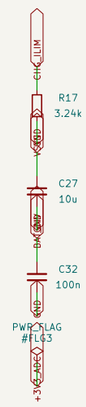












USB device only: CC1/CC2=5.1k to GND. EN2=0; EN1=GPIO23. default 100mA, 500mA after USB configuration.
Battery J2: 1=BAT+, 2=10k NTC, 3=GND. External protected 1S pack >=500mAh. No NTC means no charging.
BQ24074: ICHG=890/3570=249mA; ITERM=0.030*3300/3570=27.7mA in USB500 mode; timer nominally 6.65h.

Autonomous engineering prototype

Sheet: /Power/
File: power.kicad_sch

Title: Aster2350 — USB-C, LiPo power path and voltage monitoring

Size: A3	Date: 2026-11-09	**Rev: A**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 3/4

ESP32-C3-WROOM-02-N4

+3V3

ESP_EN
EN
3V3
IO20/RXD
IO21/TXD
RADIO_RX
RADIO_TX
IO0
IO1
ESP_STRAP2 IO2
IO3
RADIO_CTS IO4
RADIO_RTS IO5
IO6
IO7
ESP_STRAP1 IO8
ESP_BOOT IO9
IO10
IO18
IO19
GND
GND

TPS22919DCKR
U10
+3V3
IN
OUT
+3V3_RF
RADIO_POWER
ON
GND
QOD
+3V3_RF
GND

microSD
J3
SD_DAT2
SD_CS
SD_MOSI
+3V3
SD_SCK
GND
SD_MISO
SD_DAT1
GND
SD_DETECT
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET_A
DET_B
SHIELD
SH
GND

Qwiic / STEMMA QT
J4
GND
+3V3
I2C_SDA
I2C_SCL

TLV61046ADBVR
U11
+3V3
RGB_ENABLE
GND
VIN
EN
GND
SW
VOUT
FB
RGB_SW
+5V_RGB
RGB_FB

SN74AHCT1G125DBVR
U12
+5V_RGB
RGB_DATA
VCC
GND
RGB_DOUT_BUF
GND

SWD
J5
SWCLK
GND
SWDIO

GPIO A
J6
+3V3
GND
GPIO1
GPIO2
GPIO3
GPIO8
GPIO9
GPIO15

WS2812B-2020
+5V_RGB
RGB_DIN
DIN
VSS
VDD
DOUT
RGB
GND

GPIO B
J7
GND
GPIO18
GPIO19
GPIO20_ADC4
GPIO21_ADC5
GPIO22_ADC6
RUN
+3V3

RADIO_POWER
R30
100k
GND

ESP_BOOT
SW3
ESP_BOOT
GND

+3V3
C41
100n
GND
C44
4.7u
GND
+1V1
TP4
+1V1
RGB_ENABLE
TP13
RGB_ENABLE

+3V3_RF
R31
10k
ESP_EN
R35
10k
BAT_ADC
R42
2.2k
BAT_ADC
R47
100k
GND
BAT
TP5
BAT

ESP_EN
C36
100n
GND
R36
10k
GND
RADIO_RX
R43
2.2k
RADIO_RX
R48
330
BAT_ADC
TP6
BAT_ADC

+3V3
C37
1u
GND
R37
10k
GND
C42
100n
GND
C45
100n
GND
RADIO_RX
TP7
RADIO_RX

+3V3_RF
C38
47u
GND
R38
10k
GND
L2
10uH
RADIO_SW
RADIO_TX
TP8
RADIO_TX

+3V3_RF
C39
100n
GND
R39
10k
GND
R44
536k
GND
C46
100n
GND
ESP_EN
TP9
ESP_EN

+3V3_RF
R32
10k
ESP_EN
R40
3.3k
GND
R45
100k
GND
GND
TP1
GND
POWER_GOOD_N
TP10
POWER_GOOD_N

+3V3_RF
R33
10k
RADIO_RX
R41
33
RADIO_TX
R46
100k
GND
VSYS
TP2
VSYS
USB_500MA
TP11
USB_500MA

+3V3_RF
R34
10k
ESP_EN
C40
10u
GND
C43
4.7u
GND
+3V3
TP3
+3V3
RUN
TP12
RUN

GPIO4/5 = UART1 TX/RX; GPIO6/7 = UART1 CTS/RTS. ESP IO4=CTS and IO5=RTS require custom firmware pin configuration.
Radio power GPIO22 and RGB enable GPIO17 default OFF. Before radio power-off, set all four UART pins high impedance.
I2C0 GPIO20/21. microSD SPI1 GPIO10/11/12/13 + detect GPIO14. All headers and connectors are installed by PCBWay.

Autonomous engineering prototype
Sheet: /Interfaces/
File: interfaces.kicad_sch
Title: Aster2350 – Wi-Fi / BLE, microSD, I2C, RGB and development connect
Size: A3 | Date: 2026-11-09 | Rev: A
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8 | Id: 4/4